

Analysis Approach

I first analysed task completion time, audio duration, editing behaviour, and character speed to identify unusual patterns in the dataset. Based on the patterns supported by the data, I selected two warning signs that appeared most relevant for identifying potentially low quality transcription work.

Excel file:

https://docs.google.com/spreadsheets/d/1azGOGdykoz0fNEQEIZztg4aEkFjpSH2i/edit?usp=drive_link&ouid=11466945906539081673&rtpof=true&sd=true

Part I: Warning Signs of Poor Quality Transcription

1. Extremely Low Listening Time

What it tells us

A transcriber who repeatedly completes a task in much less time than the audio duration may not be listening to the complete recording. Since the task requires listening to the audio and checking the transcription, extremely fast completion can indicate rushed work.

How to measure it

I created a Listening Ratio:

$\text{Listening Ratio} = \text{Time Taken by User} \div \text{Audio Duration}$

This allowed me to compare tasks with different audio lengths fairly. The overall median Listening Ratio was 6.24 and the 25th percentile was 2.29. However, 3,579 tasks had a Listening Ratio below 0.5, meaning they were completed in less than half of the audio duration.

Red flag threshold:

I would treat a Listening Ratio below 0.5 as a warning sign, especially when this behaviour is repeated across multiple tasks.

False alarms:

A short or simple audio clip may be easy to verify quickly. An experienced transcriber may also work faster, or the platform may allow faster audio playback. Therefore, one fast task should not lead to an account block.

2. Fast Submission Without Editing Whisper Text

What it tells us

A stronger warning sign appears when a task is completed very quickly and the Whisper transcription is submitted without any changes. This may indicate that the transcriber accepted the AI output without properly checking the audio.

How to measure it

I combined two fields:

Listening Ratio below 0.5 + is_edited = No
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The analysis found 2,524 tasks with this pattern. Among the fast tasks analysed, most were submitted without editing the Whisper text. I also found that No Edit tasks had a median Listening Ratio of only 1.28, compared with 6.24 for the overall dataset.

Red flag threshold:

I would flag users who show the Fast No Edit pattern repeatedly across a minimum number of completed tasks. I would not use a single occurrence as a blocking condition.

False alarms:

Whisper may already produce a completely correct transcription, so no edits may genuinely be required. Simple audio clips can also be checked quickly. Therefore, this pattern should be combined with repeated behaviour and additional quality checks before taking action.

Part II: Automatic System to Identify Problematic Transcribers

I would not block a transcriber based on one suspicious task. The dataset showed that most users had only one recorded task, so a single task does not provide enough evidence to judge a user's overall work quality.

I would recommend tracking the **Fast No Edit pattern** for every transcriber across multiple tasks.

Recommended System

Stage 1: Task Level Flag

A task should receive a warning flag when:

Listening Ratio < 0.5 AND is_edited = No
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This means the task was completed in less than half of the audio duration and the available Whisper transcription was submitted without any changes.

The task should be flagged for monitoring, but the user should not be blocked.

Stage 2: User Level Monitoring

The platform should observe at least 10 completed tasks before making an account level decision.

For each user, calculate:

$\text{Fast No Edit Rate} = \text{Fast No Edit Tasks} \div \text{Total Completed Tasks}$
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If the pattern appears only once or twice, the user should continue normally.

If **50% or more** of the user's first 10 tasks **show the Fast No Edit pattern**, the account should be **marked as high risk** and selected transcription tasks should be sent for quality review.

Stage 3: Quality Check

For high risk users, a reviewer should **compare a sample of their user_text with the actual audio**.

If the reviewed tasks contain transcription errors, the user should receive a quality warning and temporary task restriction.

The platform can also show a message reminding the transcriber to listen to the complete audio before submitting.

Stage 4: Blocking Rule

I would block a transcriber only when:

The user repeatedly shows a Fast No Edit Rate of 50% or more across at least 10 tasks, and a manual quality review confirms inaccurate transcription in the reviewed sample.

If the same behaviour continues after a warning, the account should be blocked.

Why I Would Use This System

This system avoids blocking users only because they work quickly or because Whisper produced an accurate transcription.

My analysis also showed that most users in the provided dataset had only one recorded task. Therefore, using a single task or a 100% rate based on one task could create false positives.

The system combines behavioural data, repeated patterns, and actual transcription quality review before blocking an account. This makes the decision safer for good transcribers while still allowing the platform to identify low effort work early.